

- 1 (a) units for D identified as kg m s^{-2} M1
 all other units shown: units for A : m^2 units for v^2 : $\text{m}^2 \text{s}^{-2}$ units for ρ : kg m^{-3}
- $$C = \frac{\text{kg m s}^{-2}}{\text{kg m}^{-3} \text{m}^2 \text{m}^2 \text{s}^{-2}}$$
 with cancelling / simplification to give C no units A1 [2]
- (b) (i) straight line from (0,0) to (1,9.8) $\pm$ half a square B1 [1]
- (ii) $\frac{1}{2} mv^2 = mgh$ or using $v^2 = 2as$ C1
 $v = (2 \times 9.81 \times 1000)^{1/2} = 140 \text{ m s}^{-1}$ A1 [2]
- (c) (i) weight = drag (D) (+ upthrust) B1 [1]
 Allow mg or W for weight and D or expression for D for drag
- (ii) 1. $mg = 1.4 \times 10^{-5} \times 9.81$ C1
 $1.4 \times 10^{-5} \times 9.81 = 0.5 \times 0.6 \times 1.2 \times 7.1 \times 10^{-6} \times v^2$ M1
 $v = 7.33 \text{ m s}^{-1}$ A0 [2]
2. line from (0,0) correct curvature to a horizontal line at velocity of 7 m s^{-1} M1
 line reaches 7 m s^{-1} between 1.5 s and 3.5 s A1 [2]

- 2 (a) (resultant) force = rate of change of momentum / allow proportional to